Project Euler: Mathematical Notes & Reductions

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Problem 1: Multiples of 3 or 5

Archive Link: https://projecteuler.net/problem=1

Core Idea

Instead of iterating through all $n < N$, we use the Principle of Inclusion-Exclusion paired with closed-form arithmetic series evaluation to compute the sum in $\mathcal{O}(1)$ arithmetic operations.

Lemma. *Let $S_k(N)$ denote the sum of positive multiples of k strictly less than N. Setting $m = \lfloor \frac{N-1}{k} \rfloor$, we have:*

$$S_k(N) = \sum_{i=1}^{m} ki = k\frac{m(m+1)}{2}.$$

Derivation

By inclusion-exclusion, the sum of all multiples of 3 or 5 strictly below $N = 1000$ is given by subtracting the intersection (multiples of $\text{lcm}(3,5) = 15$):

$$\text{Total} = S_3(1000) + S_5(1000) - S_{15}(1000).$$

Problem 2: Even Fibonacci Numbers

Archive Link: https://projecteuler.net/problem=2

Core Idea

Every third Fibonacci number is even. We can derive a direct second-order linear recurrence for the subsequence of even Fibonacci numbers, eliminating the need to compute odd terms or check parity.

Proposition. *Let $F_1 = 1, F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$. Define $E_k = F_{3k}$ as the k-th even Fibonacci number. Then E_k satisfies the recurrence:*

$$E_k = 4E_{k-1} + E_{k-2} \quad \text{for } k \geq 3,$$

with seed values $E_1 = 2$ and $E_2 = 8$.

Proof. Expanding the standard recurrence relations down to indices divisible by 3:

$$\begin{aligned} F_{3k} &= F_{3k-1} + F_{3k-2} \\ &= (F_{3k-2} + F_{3k-3}) + F_{3k-2} = 2F_{3k-2} + F_{3k-3} \\ &= 2(F_{3k-3} + F_{3k-4}) + F_{3k-3} = 3F_{3k-3} + 2F_{3k-4} \\ &= 3F_{3k-3} + F_{3k-4} + (F_{3k-5} + F_{3k-6}) \\ &= 4F_{3k-3} + F_{3k-6} = 4E_{k-1} + E_{k-2}. \end{aligned}$$

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